

AI-Powered IoT Network Anomaly Detection System

1. Introduction

Modern IoT environments generate vast amounts of sensor data, making it challenging to identify abnormal device behavior manually. These anomalies may arise from hardware issues, cyberattacks, spoofed data, or malfunctioning sensors. To address this challenge, this project implements a real-time anomaly detection framework using a Raspberry Pi and an Isolation Forest machine learning model.

The system continuously monitors IoT data transmitted over MQTT and classifies each reading as normal or anomalous. Whenever an anomaly is detected, the Raspberry Pi triggers a buzzer alert, logs the event, and sends an MQTT alert notification. The framework is lightweight, scalable, and capable of running entirely on edge hardware.

A total of **3000 data samples** were collected from **synthetic IoT data sources**, generated by both **Machine-64 (laptop)** and the **Raspberry Pi**, forming the training dataset. After training, the Isolation Forest model achieved approximately **98% detection accuracy**, making it highly effective for real-time anomaly identification.

2. Components Used

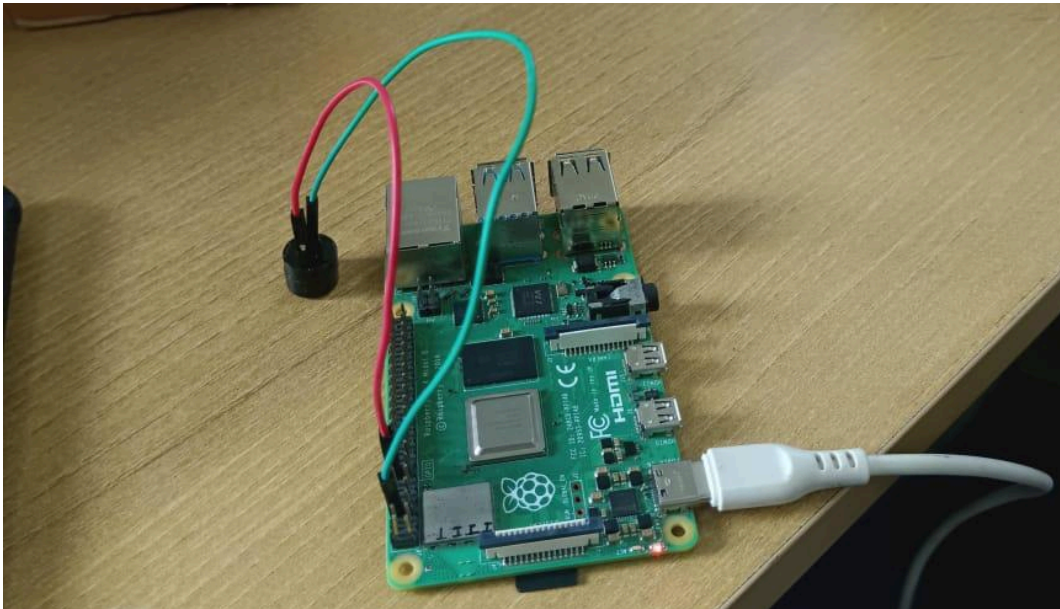
Hardware

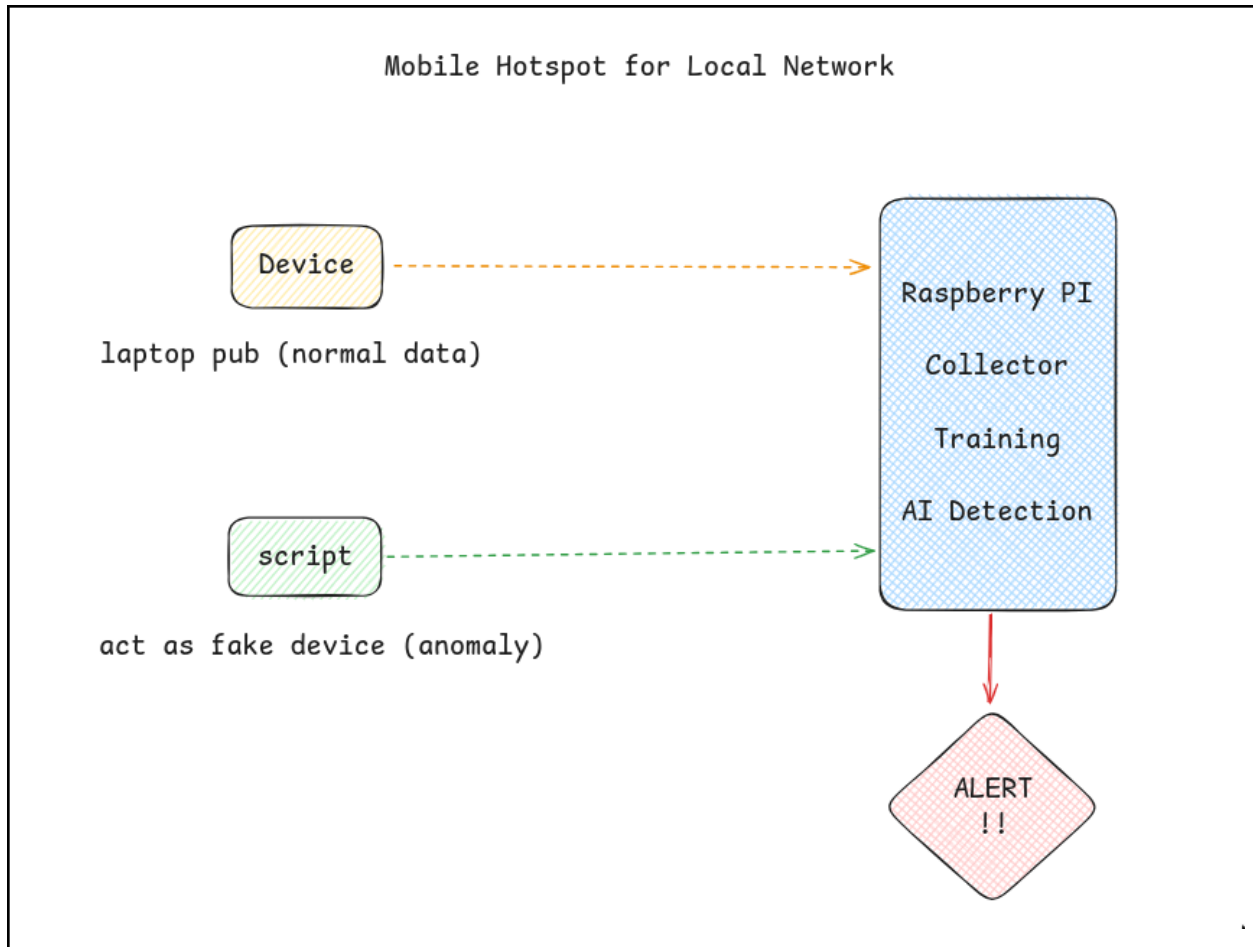
1. Raspberry Pi 4B
Acts as an AI edge device running the anomaly detection script.
2. GPIO Buzzer
Provides audible alerts when anomalies are detected.
3. Mobile Hotspot
Used as a communication bridge between the laptop and Raspberry Pi.
4. Laptop / PC
Publishes sensor data and runs training/monitoring scripts.

Software

1. Python 3
Core programming language.
2. MQTT Protocol (paho-mqtt)
Used to publish and subscribe to IoT data.
3. Scikit-Learn
For Isolation Forest anomaly detection.
4. Joblib
For saving machine learning models.
5. Pandas & NumPy
Data handling and preprocessing.
6. RPi.GPIO
Buzzer control on Raspberry Pi.

3. Circuit Diagram/Design of the circuit





Circuit Explanation

The system uses a simple alerting mechanism via a buzzer connected to the Raspberry Pi GPIO pins:

- Buzzer Positive (+) → GPIO Pin 18
- Buzzer Negative (−) → Ground (GND)

The Raspberry Pi runs `ai_anomaly.py`, which processes incoming MQTT messages. If the machine learning model identifies the data as anomalous, the script:

1. Activates the buzzer through GPIO pin 18
2. Prints an alert message on the terminal
3. Publishes an anomaly notification on the MQTT broker

System Data Pipeline

1. Data Generation

- Synthetic IoT data is generated by **machine-64** and **Raspberry Pi**.
- Data includes temperature, humidity, and motion values.

2. Data Collection

- The `collector.py` script subscribes to `devices/data` and saves all incoming data into `sensor_data.csv`.

3. Training Phase

- A total of **3000 data points** are used to train the model.
- The Isolation Forest model is trained after normalization.
- The trained model achieves **98% accuracy** in distinguishing anomalies.

4. Real-Time Detection

- The Raspberry Pi runs `ai_anomaly.py`.
- Incoming MQTT messages are processed and scaled.
- The model predicts normal or anomalous behavior.
- If anomalous:
 - Buzzer activates
 - Alert message printed in terminal
 - MQTT alert sent to `alerts/anomaly`

4. Screenshots

1. Laptop Publishing Sensor Data

- From Machine-64 Synthetic Data

```
> test ./laptop_pub.py
Published (Normal): {'device': 'Machine-64', 'temperature': 29.09705370986378, 'humidity': 42.53351458865887, 'motion': 0}
Published (Normal): {'device': 'Machine-64', 'temperature': 27.349315703529633, 'humidity': 42.84225192530477, 'motion': 0}
Published (Normal): {'device': 'Machine-64', 'temperature': 28.487504103105195, 'humidity': 40.38355947832314, 'motion': 0}
Published (Normal): {'device': 'Machine-64', 'temperature': 26.7240331622743, 'humidity': 41.027350193593904, 'motion': 0}
```

- From Raspberry PI 4b Synthetic Data

```
(venv) raspberrypi@raspberrypi:~/iot-sentinal $ ./laptop_pub.py
Published (Normal): {'device': 'raspberrypi', 'temperature': 25.0, 'humidity': 52.27, 'motion': 0}
Published (Normal): {'device': 'raspberrypi', 'temperature': 25.0, 'humidity': 52.27, 'motion': 1}
Published (Normal): {'device': 'raspberrypi', 'temperature': 25.0, 'humidity': 52.27, 'motion': 0}
Published (Normal): {'device': 'raspberrypi', 'temperature': 25.0, 'humidity': 52.27, 'motion': 0}
```

2. CSV Dataset File

timestamp	device	temperature	humidity	motion
1763531284	raspberrypi	25	52.17	1
1763531286	raspberrypi	25	52.17	0
1763531288	raspberrypi	25	52.17	0
1763531290	Machine-64	29.49265305	44.05104508	0
1763531290	raspberrypi	25.05	52.17	1
1763531291	Machine-64	27.55653148	49.36808079	0
1763531292	raspberrypi	25	52.17	0
1763531294	raspberrypi	25.02	52.17	0
1763531296	Machine-64	28.77841496	45.76373106	0
1763531297	raspberrypi	25.03	52.17	1
1763531297	Machine-64	27.58089903	40.85696909	0
1763531298	Machine-64	29.20574117	48.21700717	0
1763531299	raspberrypi	25.05	52.17	1
1763531299	Machine-64	27.30404835	42.16883098	0
1763531300	Machine-64	26.95048162	48.49666826	0
1763531301	raspberrypi	25	52.17	0

3. Model Training Output

```
(venv) raspberrypi@raspberrypi:~/iot-sential $ ./train_model.py
[TRAIN] Loading dataset...
[TRAIN] Training on 3367 rows
[TRAIN] Scaler saved
[TRAIN] Training complete. Model saved.
(venv) raspberrypi@raspberrypi:~/iot-sential $
```

4. Fake Device fake_device.py

```
> test ./fake_device.py
/home/krushang/test/./fake_device.py:10: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client()
Published (Anomalous): {'device': 'Fake-Machine-X', 'temperature': -1.0499522835723507, 'humidity': 5.804893149853951, 'motion': 1}
Published (Anomalous): {'device': 'Fake-Machine-X', 'temperature': 70.5079386508581, 'humidity': 92.58169453476296, 'motion': 1}

```

5. Raspberry Pi Running ai_anomaly.py

```
(venv) raspberrypi@raspberrypi:~/iot-sential $ ./ai_anomaly.py
Starting AI anomaly detector...
Connected to MQTT Broker
OK [Machine-64] T=29.4 H=49.2 M=0.0
OK [raspberrypi] T=25.0 H=52.7 M=0.0
OK [Machine-64] T=27.1 H=40.6 M=0.0
OK [raspberrypi] T=25.0 H=52.7 M=1.0
OK [Machine-64] T=28.5 H=41.4 M=0.0
OK [Machine-64] T=27.5 H=42.4 M=0.0

[ANOMALY ALERT] Device=Fake-Machine-X, Temp=87.50, Hum=10.19, Motion=1.0
OK [raspberrypi] T=25.2 H=52.7 M=0.0
OK [Machine-64] T=27.0 H=45.6 M=0.0
OK [raspberrypi] T=25.0 H=52.7 M=0.0
OK [Machine-64] T=27.2 H=41.7 M=0.0
OK [Machine-64] T=28.7 H=46.2 M=0.0
OK [raspberrypi] T=25.1 H=52.7 M=0.0
OK [Machine-64] T=28.1 H=49.3 M=0.0
OK [raspberrypi] T=25.0 H=52.7 M=0.0
OK [Machine-64] T=29.5 H=46.6 M=0.0

[ANOMALY ALERT] Device=Fake-Machine-X, Temp=9.09, Hum=83.36, Motion=1.0
```

5. Video & Project Folder Link

➤ https://drive.google.com/drive/folders/lpFnu39FUej_V6z8yF-QYAg4TpS82LdLV?usp=sharing

6. Conclusion

This project successfully demonstrates a practical AI-based anomaly detection system for IoT environments using Raspberry Pi, MQTT communication, and an Isolation Forest model. Using **3000 synthetic data samples** generated from both **machine-64** and the **Raspberry Pi**, the trained model achieved **98 percent accuracy**, making it reliable for identifying unusual sensor behavior.

The system performs real-time detection on the edge, triggers buzzer alerts for immediate response, and sends MQTT notifications for monitoring. Its lightweight design, low cost, and scalability make it suitable for smart homes, laboratories, and small industrial setups.

Overall, the project shows that combining IoT data with machine learning can significantly improve device security and enable rapid detection of abnormal conditions.